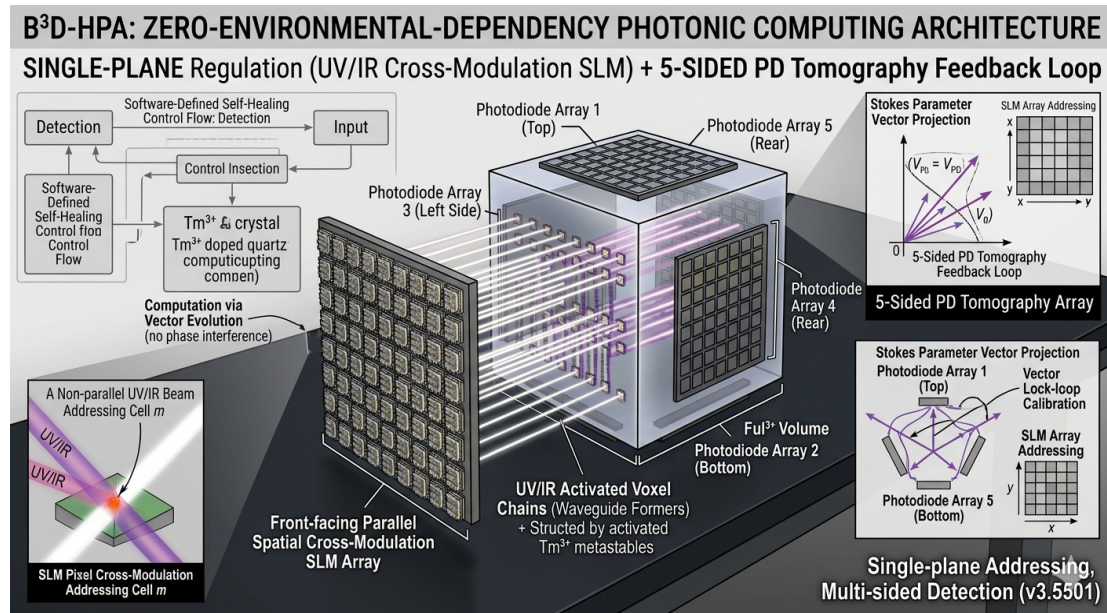


## Key Updates & Version Notes



This version incorporates the finalized system-level schematic, completing the closed-loop design from conceptual logic to engineering implementation.

### 1. Core System: Single-Plane UV/IR Spatial Cross-Modulation SLM + 5-Sided PD Array

The design diagram confirms the single-plane, side-pumping-free architecture, using only a front-facing UV/IR mixed parallel SLM array to achieve:

- 3D voxel addressing: Non-parallel UV/IR beams intersect at a focal point to form an energy threshold, activating Tm<sup>3+</sup> metastable traps precisely at the (x,y,z) voxel without path damage.
- Polarization operator writing: The IR beam induces local birefringence at the activated voxel, directly implementing polarization-based logic gates.
- 5-sided PD tomography: Top/bottom/left/right/rear photodiode arrays capture scattered and transmitted light for full-volume topology reconstruction and runtime calibration.

### 2. Polarization Incoherent Computing Paradigm

Abandons phase interference for high robustness:

- Logic states are mapped to Poincaré sphere coordinates.
- Computation is realized via UV/IR-induced local anisotropy, not phase superposition.
- Maintains high SNR even with thermal drift, eliminating the need for precision environmental control.

### 3. Multi-Method Virtual Waveguide Writing

Multiple refractive index gradient induction strategies are supported, selectable or combinable:

1. UV/IR cross-modulation (native core method)
2. Photothermal-assisted writing
3. Photo-induced refractive index (PRI) enhancement
4. Multi-wavelength IR cooperative confinement
5. Voxel-level array superposition

### 4. Software-Defined Layer: Topology Equivalence & Self-Healing Control

The core paradigm of "flexible logical topology" is formalized with three key algorithms:

- Optical Homing: SLM-to-voxel dynamic mapping via blind scanning and trilateration.
- Self-Organizing Routing: Reinforcement learning-based path formation via PD feedback.
- Vector Lock-loop: Real-time polarization correction via Stokes parameter monitoring.

### 5. Photonics HLS: From High-Level Language to Physical Collapse

Defines the complete compilation flow:

1. Human layer: Logical abstraction and Optical ISA definition.
2. Silicon layer: Topological sketching and linear mapping.
3. SLM layer: Parallel execution and runtime verification.

Supports both static curing (photonics FPGA) and streaming trigger (dynamic processor) modes, with a recommended "static skeleton + dynamic fine-tuning" hybrid mode.

### 6. Core Value & Future Directions

- Simplified, robust, software-defined, and mass-production-ready.
- Future work includes ISA development, millisecond rerouting, prototype integration, and waveguide writing method optimization.

B<sup>3</sup>D-HPV Lanthanide Quartz Photonic Computing Chip Supplementary Paper  
Memorandum: Optical Field Control and Software-Defined System for Polarization  
Incoherent Computing

### I. New Additions to the Core System: 3D Voxel Addressing and Polarization Incoherent

## Computing Based on Single-Plane Regulation

### 1. Dual-Band Cross-Modulation Architecture Without Side Pumping

Abandoning the traditional side pumping structure, this system achieves 3D voxel addressing and polarization operator writing solely through a front-side parallel infrared/ultraviolet hybrid array spatial cross laser modulation SLM, realizing path-error-free fixed-point activation via the spatial intersection of dual-band non-parallel optical fields:

- Ultraviolet addressing channel: Dual-wavelength ultraviolet optical fields are incident at non-parallel angles, forming an energy density threshold only at the intersection points to precisely excite  $\text{Tm}^{3+}$  metastable traps, lock the  $(x,y,z)$  3D voxel positions, and eliminate the need for additional side optical paths.
- Infrared computing channel: 1550 nm communication band infrared optical fields are incident synchronously, undergoing polarization modulation with activated  $\text{Tm}^{3+}$  ions at the intersection points to induce local birefringence/dichroism and directly realize physical implantation of polarization operators.
- Core advantages: Achieve the decoupling of addressing and computing through the geometric characteristics of optical fields, fundamentally eliminate path errors, simplify the optical path structure, and realize monolithic integration simultaneously.

### 2. Polarization Incoherent Computing Paradigm

Abandoning the reliance on precise phase interference, this paradigm constructs a highly robust computing logic with the polarization evolution of vector optical fields and the statistical response of medium anisotropy as the core:

- Map logical states to Poincaré sphere coordinates, with the spatial evolution of polarization states serving as the logical carrier.
- Complete computing through local birefringence/dichroism induced by UV/IR cross modulation, rather than optical wave phase superposition.
- As long as the polarization basis vectors remain stable, the signal-to-noise ratio of computing maintains an extremely high level even if the phase drifts with temperature, breaking away from the dependence on precise temperature control and anti-vibration platforms.

## II. Five-Side PD Array: Polarization Tomography and Full-Link Closed Loop

### 1. Functional Upgrade: From Signal Receiver to Polarization Tomography Scanner

The five-side PD array (upper, lower, left, right + rear) is no longer a mere signal acquisition terminal, but forms a polarization tomography system to realize 3D reconstruction of the polarization evolution of optical fields inside the quartz cube:

- The upper, lower, left and right PDs capture the Stokes parameters of side scattered light generated by polarization plane rotation.
- The rear PD captures the main signal flow.
- The silicon-based controller inversely solves the computing result of the input vector  $X$  processed by the polarization conversion matrix  $M$  of the quartz cube through the linear matrix operation  $Y = M \cdot X$ , without the need to know the specific optical path.

## 2. Real-Time Self-Calibration Closed Loop

During the logic writing process, the five-side PD array synchronously collects the energy level projection distribution of  $Tm^{3+}$  ion luminescence, realizing real-time monitoring of "write as seen":

- If the  $Tm^{3+}$  excitation at a certain voxel position is insufficient, the silicon-based system immediately fine-tunes the infrared power of the corresponding pixel of the SLM.
- Correct writing deviations in real time, solve the pain points of "uncertain writing and computing precision drift" in photonic computing, and improve the operational reliability of the architecture.

## III. Multi-Path Writing Implementation Methods for Virtual Waveguides

In this architecture, virtual waveguides are dynamic optical field confinement structures without physical etching. Their core is to realize the directional guidance of infrared computing light by inducing local refractive index gradients/anisotropic distributions inside the quartz cube through optical field modulation. Compatible with the SLM single-plane regulation and five-side PD closed-loop system, a variety of selectable/combinable implementation methods are available, each forming differentiated adaptation characteristics around the "refractive index gradient induction method, polarization sensitivity, stability/dynamics":

### 1. Local birefringence induction method via UV/IR cross modulation (native architecture adaptation)

UV+IR spatial cross optical fields output by the SLM activate  $Tm^{3+}$  metastable states at specified voxel points to induce local dielectric tensor distortion of the quartz lattice, forming a polarization-sensitive refractive index gradient that only exerts directional light guidance on infrared computing light with specific polarization states and has no confinement on non-target polarization light. Writing and addressing are completed synchronously without additional optical paths, serving as the core method with the highest engineering feasibility.

### 2. Photothermal effect-assisted method

Utilize the photothermal effect of ultraviolet light to form micro-scale temperature gradients on the specified paths inside the quartz cube, generate continuous refractive index gradients in the path area relying on the thermo-optical coefficient of quartz, and realize full-polarization light guidance confinement for infrared light. The power distribution of ultraviolet light can be regulated by the SLM to precisely control the range and gradient value of thermal gradients, adapting to scenarios requiring wide polarization compatibility, with slight thermal crosstalk that needs real-time fine-tuning in conjunction with the five-side PD array.

### 3. Photo-induced refractive index change (PRI) enhancement method

Perform trace photosensitive doping (e.g., germanium, tin) on  $\text{Tm}^{3+}$ -doped quartz, utilize the photo-induced refractive index change effect of ultraviolet light to make the quartz matrix on the cross optical field path produce permanent/semi-permanent refractive index enhancement, forming stable virtual waveguides. Suitable for photonic FPGA scenarios requiring long-term curing of logical topologies, with relatively high reconstruction difficulty after writing that requires higher-power ultraviolet light for erasure.

### 4. Multi-wavelength infrared cooperative confinement method

Without ultraviolet addressing light, directly output multiple phase-matched infrared light beams through the SLM to form interference-type refractive index gradients by utilizing the transient refractive index change of the optical Kerr effect, and quickly construct dynamic virtual waveguides. Featuring an extremely fast writing speed, it is suitable for dynamic processor scenarios with "streaming trigger", requiring slightly higher infrared light power, with a short waveguide maintenance time that needs continuous optical field refreshment.

### 5. Voxel-level array superposition method

Decompose virtual waveguides into multiple continuous 3D voxel points, activate voxel points one by one through the SLM parallel array to form a discrete refractive index gradient array, and realize continuous path light guidance through directional scattering/refraction of infrared light between adjacent voxel points. Adapting to the construction of virtual waveguides with complex 3D topologies, it can flexibly reconstruct paths by turning off individual voxel points, serving as the core method for realizing "logical rerouting" and highly compatible with self-healing algorithms.

A combined strategy can be adopted in actual implementation, such as curing the core topology via UV/IR cross modulation and fine-tuning dynamic paths via the photothermal effect, which balances the stability and reconfigurability of the system and conforms to the core design requirements of flexible topology and self-healing.

## IV. Software Control Layer: Topological Equivalence and Self-Healing Reconfigurable Photonic Computing

### 1. Core Paradigm: Logical Topological Equivalence

The SLM array writes not fixed addresses but flexible logical topologies:

- As long as the logical transformation function  $f(x)$  between input and output is correct, the specific optical paths of light inside the cube (straight line, detour, multi-voxel relay) have logical topological equivalence in physics.
- This "roughly arbitrary" guidance paradigm brings three-dimensional dimensionality reduction to engineering implementation:
- Physical layer: Fault tolerance and dynamic avoidance, capable of bypassing defective "bad points" and adjusting paths dynamically.
- Compilation layer: From "full routing" to "constraint solving", only the angle combination with the lowest energy consumption and least crosstalk needs to be found.
- Operation layer: Spatial compression of high-dimensional information, multiple logical skeletons can be written overlapping in the same space to realize multi-engine parallel operation.

### 2. Three-Level Self-Healing Control Algorithm

#### (1) Spatial Addressing Self-Calibration Program: Optical Homing

Establish a dynamic mapping relationship between the 2D pixel coordinates  $(x,y)$  of the SLM and the 3D voxels  $(X,Y,Z)$  inside the cube:

- Blind scanning: Randomly select SLM array points and emit low-power UV+IR detection pulses.
- Centroid inversion: Calculate the physical centroid of the intersection point by the trilateration method through the signal intensity difference of the five-side PD array.
- Error modeling: Establish a deviation function  $E(X,Y,Z)$  between the physical centroid and the SLM command  $(\theta,\varphi)$  to form a calibration table.

#### (2) Virtual Waveguide Enhancement Program: Self-Organizing Routing

Convert the logical sketch into a "logical manifold" inside the cube and realize path self-organization by utilizing reinforcement learning:

- Preset path: The SLM emits UV/IR hybrid light at a probability-optimal angle according to the logical sketch.
- Multi-directional feedback: If the polarization projection received by the PD is close to the target, increase the IR duty cycle of the corresponding SLM pixel to reinforce the path.
- Physical collapse: Utilize the  $Tm^{3+}$  stimulated saturation effect; the path that first forms

a logical closed loop will automatically suppress stray paths through nonlinear gain, forming a logical manifold with the least resistance and highest polarization fidelity.

### (3) Runtime Polarization Correction Program: Vector Lock-loop

Aim at the stability of in-memory computing, correct the polarization deviation caused by thermal drift and refractive index inhomogeneity in real time:

- Feature extraction: The PD captures the current polarization rotation angle  $\psi_{\text{real}}$ .
- Deviation comparison: Compare  $\psi_{\text{real}}$  with the target  $\psi_{\text{target}}$  to calculate the error.
- Dynamic compensation: Without changing the path, only fine-tune the initial polarization angle of the SLM to inject a "reverse polarization deviation" to offset the unintended birefringence inside the medium, realizing active polarization locking.

## V. Photonic High-Level Synthesis (Photonics HLS): From High-Level Language to Physical Collapse

### 1. Core of the Paradigm: Connecting Photonic Hardware and Software Definition

Map the mature logical architecture of the silicon-based world to the physical medium of quartz through a set of "optical instruction sets" as a bridge, realizing high-level synthesis in the field of photonic computing:

- Humans are no longer "circuit repairers" but "architecture designers", focusing on logical abstraction and instruction set definition.
- The silicon-based system acts as an offline compiler, responsible for topological sketching and linearization.
- The SLM compiler is responsible for physical layer writing and verification, realizing a complete compilation flow from high-level language to physical collapse.

### 2. Complete Compilation Division Flow

#### (1) Human Layer: Semantic Extraction and Optical Instruction Set Definition

- Logical abstraction: Extract large model operators (e.g., Attention, Softmax, MLP) from high-level programming languages.
- Instruction translation: Develop a set of Optical Instruction Set Architecture (Optical ISA), such as `ROTATE_POL(P1, 90)`, `VECTOR_DOT(V1, V2)`.
- High-dimensional compression: Utilize the multiple degrees of freedom of light (wavelength, polarization, spatial position) to compress thousands of lines of silicon-based binary instructions into a single "high-dimensional optical instruction packet".

#### (2) Silicon-Based Layer: Topological Sketching and Linearization

- Offline compiler: Generate a 3D logical topological sketch in the simulated space according to the optical instruction set.
- Linear mapping: Simplify the 3D optical field path into a linear mapping of "input surface coordinate → logical voxel point → polarization transformation point → output surface projection".
- SLM bitstream generation: Output an executable "phase/intensity/polarization" configuration table.

### (3) SLM Compiler Layer: Physical Layer Writing and Verification

- Parallel execution: The SLM receives the configuration table, emits UV/IR hybrid light, and induces local birefringence.
- Runtime check: The five-side PD array verifies the polarization projection results in real time.
- Dynamic correction: If the projection is inconsistent, the compiler fine-tunes the power or wavefront of the SLM in real time to maintain the correctness of the logical topology.

### 3. Mode Selection: Static Curing vs. Streaming Trigger

- Static curing (Programmable Photonic FPGA): Write all logic gates first, then let the signal light pass through; it has low requirements for SLM refresh bandwidth and is suitable for fixed inference tasks.
- Streaming trigger (Dynamic Processor): Adjust the topology while computing with high flexibility, suitable for dynamic models and online learning.
- Compromise solution: Adopt the "static skeleton + dynamic fine-tuning" mode, where core operators are cured and only polarization angle fine-tuning is performed during runtime to compensate for deviations, balancing performance and flexibility.

## VI. Core Value and Future Research Directions of the New System

### 1. Core Value

- Architecture simplification: Single-plane SLM regulation without side pumping, realizing monolithic integration of optical paths.
- Robustness improvement: Polarization incoherent computing combined with self-healing control, breaking away from the dependence on precise environments.
- Software definition: Photonics HLS paradigm greatly reduces the development threshold and adapts to high-dimensional computing scenarios such as large models.
- Engineering adaptability: The full link reuses mature processes, featuring low cost and mass production feasibility.
- Flexible expansion: Multiple virtual waveguide writing methods support scenario-based customization and combination, adapting to implementation scenarios with different computing power requirements.



## 2. Future Research Directions

- Develop a dedicated optical instruction set for large model operators to improve compilation efficiency and execution density.
- Realize millisecond-level "logical rerouting" to endow the quartz cube with the evolutionary ability to adapt to algorithm evolution.
- Complete the integrated integration of SLM, quartz cube, five-side PD array and silicon-based controller, build a prototype and conduct performance testing and scenario verification.
- Carry out optimization research on the combination of multiple virtual waveguide writing methods, and establish a writing method selection and parameter regulation system for different scenarios.
- Optimize the photosensitive/thermo-optical characteristics of  $\text{Tm}^{3+}$ -doped quartz to improve the induction efficiency and stability of virtual waveguides.